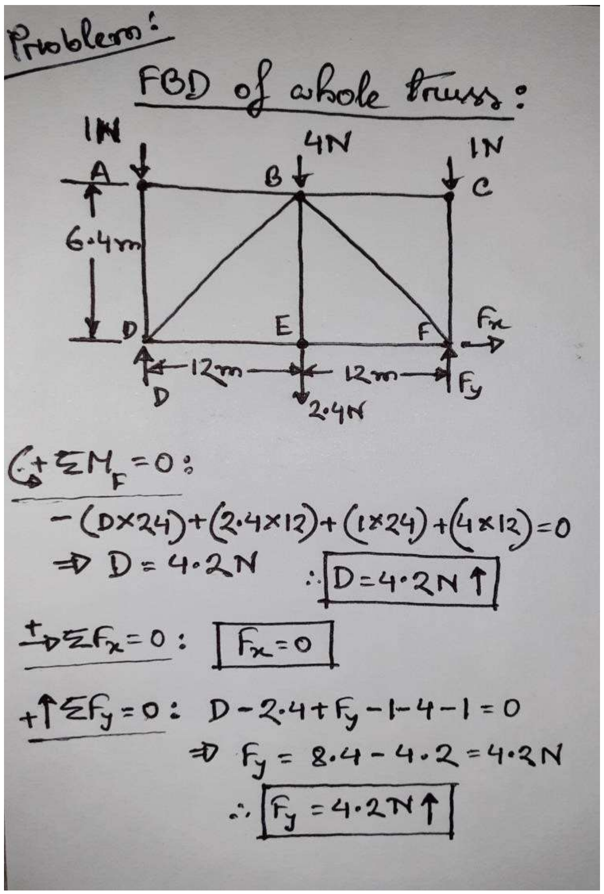
Analysis of Trusses by Method of Joints

➢ **Solution:**



Problem:

FBD of whole truss:

$\circlearrowleft + \sum M_F = 0$:

$-(D\times24)+(2.4\times12)+(1\times24)+(4\times12)=0$

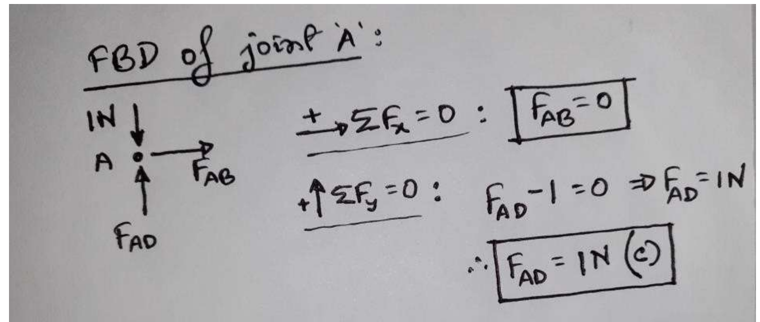
$\Rightarrow D = 4.2N$ ∴ $\boxed{D=4.2N\uparrow}$

$\xrightarrow{+}\sum F_x = 0$: $\boxed{F_x=0}$

$+\uparrow\sum F_y=0$: $D-2.4+F_y-1-4-1=0$

$\Rightarrow F_y = 8.4-4.2=4.2N$

∴ $\boxed{F_y=4.2N\uparrow}$

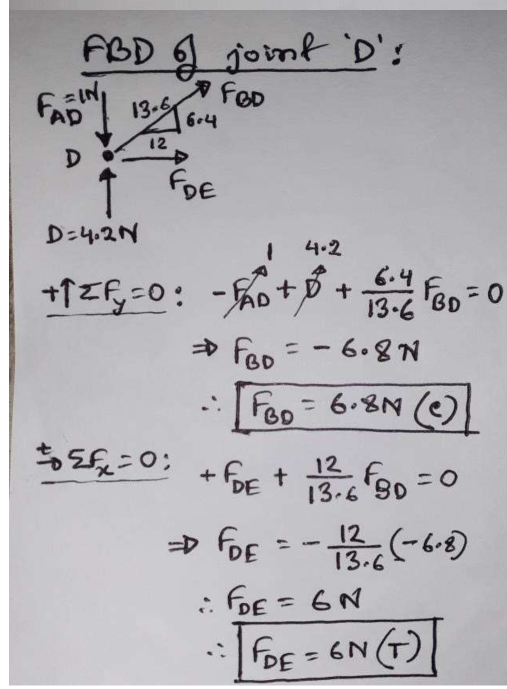


FBD of joint 'A':

$\xrightarrow{+}\sum F_x=0$: $\boxed{F_{AB}=0}$

$+\uparrow\sum F_y=0$: $F_{AD}-1=0 \Rightarrow F_{AD}=1N$

∴ $\boxed{F_{AD}=1N\ (C)}$



FBD of joint 'D':

$+\uparrow\sum F_y=0$: $-\overset{1}{\cancel{F_{AD}}}+\overset{4.2}{\cancel{D}}+\frac{6.4}{13.6}F_{BD}=0$

$\Rightarrow F_{BD}=-6.8N$

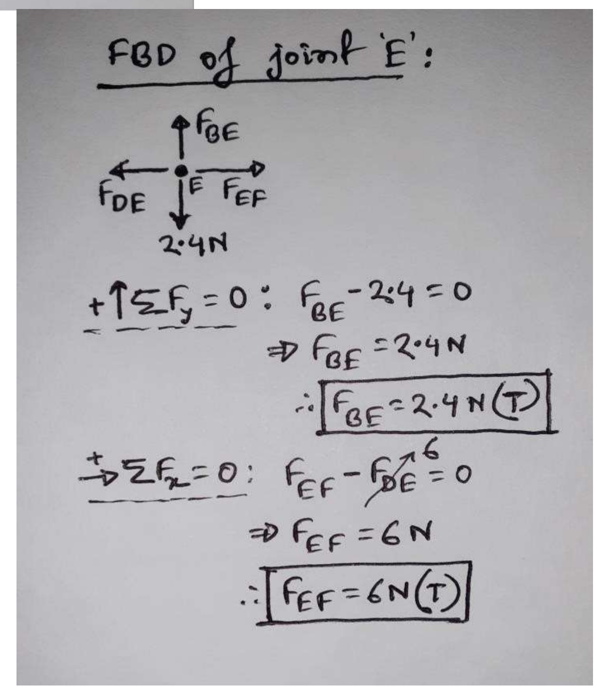
∴ $\boxed{F_{BD}=6.8N\ (C)}$

$\xrightarrow{+}\sum F_x=0$: $+F_{DE}+\frac{12}{13.6}F_{BD}=0$

$\Rightarrow F_{DE}=-\frac{12}{13.6}(-6.8)$

∴ $F_{DE}=6N$

∴ $\boxed{F_{DE}=6N\ (T)}$



FBD of joint 'E':

$+\uparrow\sum F_y=0$: $F_{BE}-2.4=0$

$\Rightarrow F_{BE}=2.4N$

∴ $\boxed{F_{BE}=2.4N\ (T)}$

$\xrightarrow{+}\sum F_x=0$: $F_{EF}-\overset{6}{\cancel{F_{DE}}}=0$

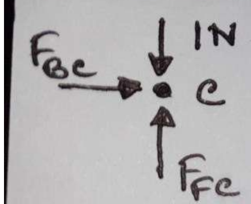
$\Rightarrow F_{EF}=6N$

∴ $\boxed{F_{EF}=6N\ (T)}$

Analysis of Trusses by Method of Joints

➢ **Solution:**

FBD of joint 'C':



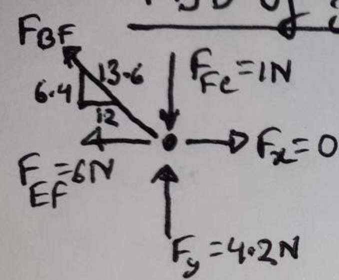
$\xrightarrow{+} \Sigma F_x = 0$: $\boxed{F_{BC} = 0}$

$+\uparrow \Sigma F_y = 0$: $F_{FC} - 1 = 0$

$\Rightarrow F_{FC} = 1N$

$\therefore \boxed{F_{FC} = 1N\,(C)}$

FBD of joint 'F':



$\xrightarrow{+} \Sigma F_x = 0$:

$\cancel{F_x}^{0} - \cancel{F_{EF}}^{6} - \frac{12}{13.6} F_{BF} = 0$

$\Rightarrow F_{BF} = -6.8N$

$\therefore \boxed{F_{BF} = 6.8N\,(C)}$

Summary of results:

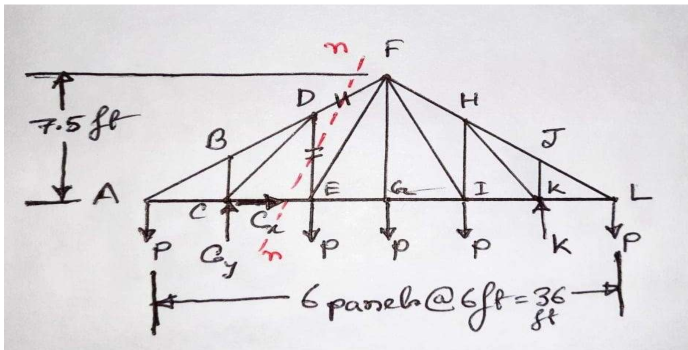
$F_{AB} = 0 = F_{BC}$; $F_{AD} = 1N\,(C) = F_{FC}$;

$F_{DE} = 6N\,(T) = F_{EF}$; $F_{BD} = 6.8N\,(C) = F_{BF}$;

$F_{BE} = 2.4N\,(T)$;

Analysis of Trusses by Method of Sections

➢ **Solution:**

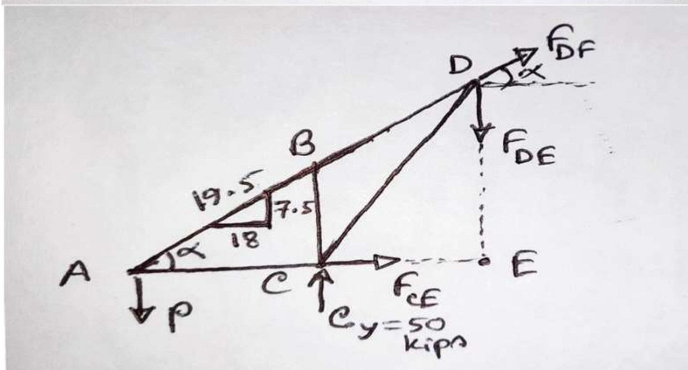


$\circlearrowright + \Sigma M_K = 0:$

$$(P\times 30) - (C_y \times 24) + (P\times 18) + (P\times 12) + (P\times 6) - (P\times 6) = 0$$

$$\Rightarrow \boxed{C_y = 50 \text{ kips}} \uparrow$$

$\xrightarrow{+} \Sigma F_x = 0: \quad \boxed{C_x = 0}$



ΔADE and ΔAFG similar;

$$\frac{DE}{12} = \frac{7.5}{18} \Rightarrow \underline{DE = 5}$$

$\circlearrowright + \Sigma M_D = 0: \quad (P\times 12) - (50\times 6) + (F_{CE}\times 5) = 0$

$$\Rightarrow \boxed{F_{CE} = 12 \text{ kips (T)}}$$

$\xrightarrow{+} \Sigma F_x = 0: \quad F_{CE} + F_{DF}\times \frac{18}{19.5} = 0$

$$\Rightarrow F_{DF} = -13 \text{ kips}$$

$$\therefore \boxed{F_{DF} = 13 \text{ kips (C)}}$$

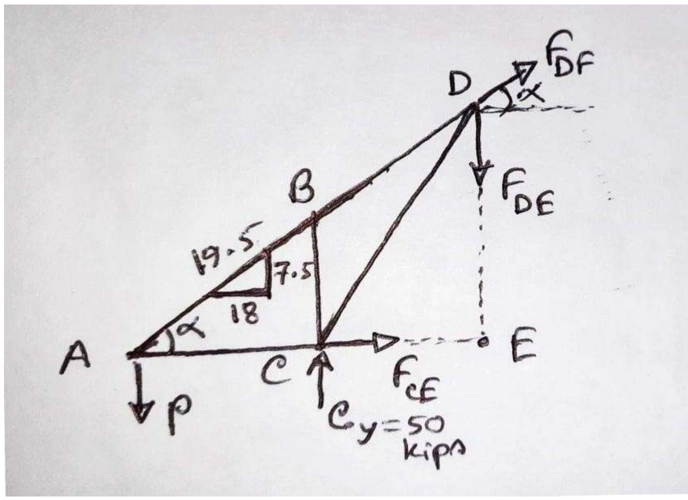
$+\uparrow \Sigma F_y = 0: \quad -P + 50 - F_{DE} + F_{DF}\times \frac{7.5}{19.5} = 0$

$$\Rightarrow F_{DE} = 25 \text{ kips}$$

$$\therefore \boxed{F_{DE} = 25 \text{ kips (T)}}$$

Analysis of Trusses by Method of Sections

➤ **Solution:**



ΔADE and ΔAFG similar;

$$\frac{DE}{12} = \frac{7.5}{18} \Rightarrow \underline{DE = 5}$$

$\circlearrowleft + \Sigma M_A = 0$:

$$+(50 \times 6) - (F_{DE} \times 12) = 0$$

$$\Rightarrow F_{DE} = 25 \text{ kips}$$

$$\therefore \boxed{F_{DE} = 25 \text{ kips (T)}}$$

$\circlearrowleft + \Sigma M_E = 0$:

$$(P \times 12) - (50 \times 6) - \left\{\left(F_{DF} \times \frac{18}{19.5}\right) \times 5\right\} = 0$$

$$\Rightarrow F_{DF} = -0.65P = -13 \text{ kips}$$

$$\therefore \boxed{F_{DF} = 13 \text{ kips (C)}}$$